Practical Telemedicine for Veterans with Persistently Poor Diabetes Control: A Randomized Pilot Trial

Matthew J. Crowley, MD,[1,2] David Edelman, MD,[1,3]
Ann T. McAndrew, RN,[4] Susan Kistler, RN,[4] Susanne Danus, BS,[1]
Jason A. Webb, MD,[3,5,6] Joseph Zanga, MD,[3,5,6]
Linda L. Sanders, MPH,[3] Cynthia J. Coffman, PhD,[1,7]
George L. Jackson, PhD,[1,3] and Hayden B. Bosworth, PhD[1,3]

[1]Center for Health Services Research in Primary Care, [4]Home Telehealth Program, and [6]Mental Health Service, Durham Veterans Affairs Medical Center, Durham, North Carolina.
Divisions of [2]Endocrinology, Diabetes, and Metabolism and [3]General Internal Medicine, Department of Medicine, Duke University, Durham, North Carolina.
[5]Department of Psychiatry and Behavioral Medicine, Duke University, Durham, North Carolina.
[7]Department of Biostatistics and Bioinformatics, Duke University Medical Center, Durham, North Carolina.

This study is registered at ClinicalTrials.gov with clinical trial registration number NCT01778751.
Portions of these data were presented at the American Diabetes Association Scientific Sessions, held in Boston, Massachusetts, June 8, 2015, and the Veterans Affairs Health Services Research and Development National Meeting, held in Philadelphia, PA, July 10, 2015.
The views expressed in this article are those of the authors and do not necessarily reflect the position of the Department of Veterans Affairs or Duke University.

Abstract

Background: *Telemedicine-based diabetes management improves outcomes versus clinic care but is seldom implemented by healthcare systems. In order to advance telemedicine-based management as a practical option for veterans with persistent poorly controlled diabetes mellitus (PPDM) despite clinic-based care, we evaluated a comprehensive telemedicine intervention that we specifically designed for delivery using existing Veterans Health Administration (VHA) clinical staffing and equipment.* ***Materials and Methods:*** *We conducted a 6-month randomized trial among 50 veterans with PPDM; all maintained hemoglobin A1c (HbA1c) levels continuously >9.0% for >1 year despite clinic-based management. Participants received usual care or a telemedicine intervention combining telemonitoring, medication management, self-management support, and depression management; existing VHA clinical staff delivered the intervention. Using linear mixed models, we examined HbA1c, diabetes self-care (measured by the Self-Care Inventory–Revised questionnaire), depression, and blood pressure.* ***Results:*** *At baseline, the model-estimated common HbA1c intercept was 10.5%. By 6 months, estimated HbA1c had improved by 1.3% for intervention participants and 0.3% for usual care (estimated difference, −1.0%, 95% confidence interval [CI], −2.0%, 0.0%;* p = *0.050). Intervention participants' diabetes self-care (estimated difference, 7.0; 95% CI, 0.1, 14.0;* p = *0.047), systolic blood pressure (−7.7 mm Hg; 95% CI, −14.8, −0.6;* p = *0.035), and diastolic blood pressure (−5.6 mm Hg; 95% CI, −9.9, −1.2;* p = *0.013) were improved versus usual care by 6 months. Depressive symptoms were similar between groups.* ***Conclusions:*** *A comprehensive telemedicine intervention improved outcomes among veterans with PPDM despite clinic-based care. Because we specifically designed this intervention with scalability in mind, it may represent a practical, real-world strategy to reduce the burden of poor diabetes control among veterans.*

Key words: *telemedicine, telehealth, e-health, home health monitoring*

Introduction

The devastating complications and costs of diabetes rise exponentially as hemoglobin A1c (HbA1c) increases.[1–3] Individuals with HbA1c >9.0%—approximately 13% of diabetes patients—are at particularly high risk for complications and costs.[4] Because improved control can reduce diabetes complications,[5] patients with persistent poorly controlled diabetes mellitus (PPDM), or maintenance of an HbA1c >9.0% for >1 year, represent a high-priority target for intervention.

Patients with PPDM often remain poorly controlled despite receiving clinic-based diabetes care with primary care providers or endocrinologists. PPDM is difficult to manage in the clinic because many factors that underlie poor control, including unreliable or unavailable blood glucose data, medication nonadherence, complex treatment regimens, and comorbid depression,[6–8] are difficult to address with the infrequent patient–provider contact achievable in the clinic setting.[9] When clinic-based care fails to resolve PPDM, approaches are needed that can better counter factors contributing to poor control.

 DOI: 10.1089/tmj.2015.0145

Telemedicine, or use of information exchanged via electronic communications for medical purposes,[10] may enhance diabetes outcomes versus clinic-based management. In studies, telemedicine approaches that address specific factors underlying poor diabetes control have proven superior to usual care. Examples of such successful telemedicine-based strategies include (1) telemonitoring to facilitate blood glucose data collection,[11] (2) self-management support to improve medication adherence,[12] (3) medication management for complex diabetes treatment regimens,[13] and (4) management for comorbid depression.[14] Given these data, a comprehensive, telemedicine-based intervention that addresses multiple factors underlying PPDM could potentially lower HbA1c levels in this high-risk, clinic-refractory group.

However, although some organizations use telemonitoring for diabetes,[15] healthcare systems have rarely integrated comprehensive telemedicine interventions into practice, even for PPDM.[16] Some of this gap stems from the fact that many studied approaches lack *scalability*—that is, they are not designed to broadly and effectively reach the target population.[17] In order for telemedicine to fulfill its potential as a means to reduce the burden of PPDM, interventions must be developed with a focus on scalability, such that effective implementation under real-world conditions is achievable.[18] Without such practical telemedicine options, clinicians will continue with little recourse when clinic-based diabetes care proves insufficient for PPDM.

In order to foster telemedicine-based management of clinic-refractory PPDM as a practical option within the Veterans Health Administration (VHA), the largest integrated healthcare system in the United States,[19] we developed Advanced Comprehensive Diabetes Care (ACDC). ACDC bundles telemonitoring, self-management support, medication management, and depression management—four evidence-based approaches that target factors underlying PPDM—into a comprehensive telemedicine intervention specifically designed for delivery using existing VHA clinical staffing and equipment. We constructed this approach to be (1) potent enough to reduce HbA1c levels in clinic-refractory PPDM and (2) amenable to practical delivery using existing VHA clinical resources, enhancing its potential scalability.

Materials and Methods

We conducted a pilot trial to evaluate the feasibility and effectiveness of using existing VHA clinical staffing and equipment to deliver ACDC for PPDM. We randomized 50 veterans with clinic-refractory PPDM to receive ACDC-augmented care or usual care for 6 months. This study was approved by the Durham Veterans Affairs Medical Center (VAMC) Institutional Review Board.

STUDY POPULATION AND ENROLLMENT

Eligible veterans had type 2 diabetes with maintenance of HbA1c continuously >9.0% for >1 year, a minimum of two HbA1c measurements during this period, and an assigned Durham VAMC primary care provider; the presence of these three criteria indicated clinic-refractory PPDM. Given challenges initiating insulin via telephone, established insulin use was also required for inclusion. Exclusion criteria included inability to communicate by telephone (e.g., no access, hearing/speech impediment), dementia, psychosis, life-limiting illness, recent cardiovascular event or stroke, active alcohol/substance abuse, prior hypoglycemic seizure or coma, refusal to perform self-monitoring of blood glucose (SMBG), or use of continuous subcutaneous insulin infusion pumps.

A research assistant sent recruitment letters to potentially eligible veterans, followed up by telephone, conducted a baseline interview, and obtained written, informed consent. Using a computer-generated, blocked sequence, a research assistant randomized consenting veterans in an unblinded fashion. Participants in both study arms attended 3-month and 6-month outcome visits, during which they were interviewed and underwent phlebotomy for HbA1c measurement. Veterans were not compensated for study participation.

ACDC INTERVENTION

We partnered with Durham VHA Home Telehealth (HT) program nurses to create ACDC. VHA has implemented HT programs nationwide, for which all veterans with poor diabetes control qualify.[15] Although HT nurses regularly telemonitor diabetes, they do not currently deliver comprehensive diabetes management interventions like ACDC. ACDC's delivery relies on existing VHA HT staffing and equipment; repurposing ubiquitous HT infrastructure in this manner creates the potential for scaling throughout VHA.

Veterans randomized to ACDC-augmented care were contacted by one of two study-affiliated HT nurses (A.T.M. or S.K.), who provided the standard Durham VAMC HT orientation and standard telemonitoring equipment (TELERESPONSE™ interactive voice response system; Cardiocom®, Chanhassen, MN). As would be the case in real-world practice, HT implemented the intervention; research staff managed only study-specific tasks (e.g., recruitment, randomization, outcome ascertainment). HT delivered ACDC's four components during scheduled telephone encounters every 2 weeks for 6 months and documented encounters in the VHA electronic health record (EHR).

Telemonitoring. Participants were asked to perform SMBG before meals and at bedtime and instructed to transmit these data on a daily basis using their HT-issued equipment. Per local

HT policy, participants received daily automated calls to prompt data transmission; upon receiving a call, veterans reported their SMBG data using the interactive voice response system or their telephone's number keys. Failure to submit data for 3 days triggered a call from HT, followed by a letter at 5 days. At 2-week intervals, HT contacted intervention participants by telephone to review submitted SMBG data, reconcile medications, and assess diabetes medication adherence.

Self-management support. During most 2-week telephone encounters, HT delivered a diabetes self-management support module. Modules were adapted for ACDC from a prior study[20]; the eight topics were SMBG, hypoglycemia recognition and management, diabetes medication use, diet in diabetes (two modules), self-managing insulin, exercise in diabetes, and diabetes complications. Participants were given the option to repeat a module of their choice at two encounters.

Physician-guided medication management. Following each 2-week telephone encounter, HT compiled an EHR-based report summarizing SMBG data, reconciled medications, adherence, and other relevant information. HT then forwarded this report via EHR to the study physician (M.J.C.). Medication changes were at the discretion of the study physician; insulin titration based on SMBG was the primary emphasis, with adjustment of oral antihyperglycemic agents as indicated. Recommendations were transmitted via EHR back to HT, whose nurse then relayed changes to veterans by telephone. HT documented medication changes in the EHR, alerting primary care providers to all notes. HbA1c goals were individualized as per American Diabetes Association guidelines[21]; in order to minimize hypoglycemia, <8.0% was the typical target.

Physician-guided depression management. If veterans endorsed depressive symptoms at baseline or 3 months, HT asked the study psychiatrist (J.A.W. or J.Z.) to call and arrange an in-person evaluation within 7–14 days. Our approach emphasized pharmacotherapy proven effective in diabetes populations.[22] If participants endorsed suicidal ideation during an outcome ascertainment interview, they received emergency assessment at the Durham VAMC. In cases where a psychiatry provider was already actively managing a veteran's depression, we deferred to the existing treatment plan.

USUAL CARE

Participants randomized to usual care were not contacted by HT but received an educational packet and continued diabetes management with existing providers.

STUDY MEASURES

Our primary effectiveness outcome was HbA1c measured at baseline and 3 and 6 months. Secondary outcomes were diabetes self-care (Self-Care Inventory–Revised),[23] depressive symptoms (Patient Health Questionnaire-9),[24] and self-reported medication adherence (Morisky Medication Adherence Scale-4, dichotomized using a validated approach).[25,26] We also examined blood pressure (BP); participants' BP was measured three times at each outcome assessment and averaged.

We ascertained adverse events through structured self-report and EHR review.[27] Because hypoglycemia was the primary potential adverse event, we carefully analyzed the ACDC group's HT-reported SMBG data for blood glucose values <70 mg/dL and <60 mg/dL. Because usual care participants did not transmit SMBG data as part of the study, we did not analyze these data for this group.

Finally, in order to help refine ACDC and its delivery strategy, we tracked telephone encounters and nurse call times and assessed intervention acceptability at 6 months using the validated treatment acceptability and preferences (TAP) measure.[28] We also conducted semistructured interviews with intervention participants to explore perceptions about ACDC (qualitative analysis is ongoing so data are not presented).

SAMPLE SIZE

Our main objective was to evaluate the feasibility of using existing VHA infrastructure to deliver ACDC; we chose our sample size of 50 veterans with this goal in mind. Because we also wished to assess ACDC's effectiveness, we performed power calculations for HbA1c based on our sample size. Assuming a standard deviation (SD) of 1.5% and a loss to follow-up of 20%,[29] we conservatively estimated that 50 participants would yield 80% power to detect a 1.4% HbA1c between-group difference (two-sided alpha = 0.05).[30]

ANALYSIS

All participants were analyzed in their assigned randomization group on an intention-to-treat basis. We examined baseline demographic/clinical data using descriptive statistics. We used linear mixed models with a constrained intercept and unstructured covariance to analyze the primary outcome and continuous secondary outcomes (diabetes self-care, depressive symptoms, BP).[31,32] We used all available data; no observations were deleted for missing follow-up. The estimation procedure used in our longitudinal analysis framework yields unbiased estimates of parameters when missing outcomes are assumed to be ignorable (i.e., related to observed covariates or response variables but not unobserved variables).[33] The primary model predictors included a common intercept, indicator variables for

 © MARY ANN LIEBERT, INC.

follow-up time, and treatment arm by follow-up time indicator interaction variables. We evaluated self-reported medication adherence as a binary outcome using a generalized estimating equation with a logit link function and an unstructured correlation structure.[34]

We descriptively examined telephone encounters, nurse call times, and intervention acceptability. We descriptively characterized adverse events, including self-reported hypoglycemia and, for intervention participants, rates of hypoglycemia on HT-collected SMBG data (usual care participants did not transmit SMBG).

Results

POPULATION CHARACTERISTICS

Participants were enrolled from December 2013 to April 2014 and followed up through December 2014. The population was predominantly male and majority African American, with a mean age of 60 years and poor baseline diabetes control. Randomization groups were generally well balanced (*Table 1*). Observed baseline HbA1c was 10.2% in the ACDC group and 10.7% for usual care, with a linear mixed models–estimated common intercept of 10.5%. Forty-six of 50 veterans ($n = 23$ in each group) completed their 6-month assessment (*Fig. 1*).

STUDY OUTCOMES

At 3 months, mean HbA1c had improved by 1.3% for the ACDC group and 0.3% for usual care (estimated difference, −1.0%, 95% confidence interval [CI], −1.7 to −0.2; $p = 0.012$) (*Table 2*). A −1.0% difference was also seen at 6 months (95% CI, −2.0 to −0.0; $p = 0.050$). Although the intervention did not address hypertension, ACDC participants had significantly lower systolic (−7.7 mm Hg; $p = 0.035$) and diastolic (−5.6 mm Hg; $p = 0.013$) BP at 6 months versus usual care (*Table 2*).

ACDC participants had significantly better diabetes self-care at 6 months than usual care (estimated difference in Self-Care Inventory–Revised = 7.0; $p = 0.047$), but self-reported medication adherence and depressive symptoms did not differ between groups (*Table 2*). It is notable that of the seven ACDC participants with depressive symptoms at study baseline, six already had VHA psychiatry care; consequently, only one veteran required expedited study psychiatrist evaluation.

The ACDC group completed an average of 8.7 (SD 4.8) of the 12 scheduled telephone encounters. Of the 23 ACDC participants with 6-month follow-up data, 16 completed six or more encounters (*Fig. 1*). These "engaged" veterans had a particularly strong response to ACDC, with an observed mean HbA1c improvement from 10.2% at baseline to 8.3% at 6 months (*Table 3*). The remaining seven intervention participants completed three or fewer telephone encounters and had an observed mean HbA1c of 10.8% at 6 months, versus 10.3% at baseline (*Table 3*).

Nurses spent approximately 30 min conducting each intervention telephone call. Although not formally tracked for the purposes of this pilot, estimated study physician time was

Table 1. Study Population Baseline Data, Stratified by Intervention Arm

BASELINE CHARACTERISTICS[a]	ACDC ($N = 25$)	USUAL CARE ($N = 25$)
Demographics		
Age (years) [mean (SD)]	60 (8.4)	60 (9.2)
Male	25	23
Race		
White	13	8
African American	12	15
Other	0	2
Latino/Hispanic ethnicity	1	2
Married/living together	18	13
Completed ≤12 years of school	9	10
Low health literacy (REALM ≤60)	11	9
Employed, full-time/part-time/self	10	8
Inadequate income[b]	5	8
Lives alone	6	5
Clinical data		
Duration of diabetes in years [median (IQR)]	12 (13)	12 (9)
HbA1c (%) [mean (SD)]	10.2 (1.4)	10.7 (1.6)
SBP (mm Hg) [mean (SD)]	127 (16.4)	127 (16.7)
DBP (mm Hg) [mean (SD)]	80 (12.6)	80 (10.1)
Diabetes self-care (SCI-R) [mean (SD)]	52 (14.8)	50 (9.6)
Active depressive symptoms (PHQ-9 ≥ 10)	7	5
Self-reported medication nonadherence[c]	15	17

[a]Unless otherwise noted, number with characteristic is given.

[b]Inadequate income refers to patients who can pay bills only after cutting back expenditures or have difficulty paying bills despite cutting back expenditures.

[c]Using a 4-point Likert scale, patients addressed the extent to which four statements applied to their medication-taking behavior during the past 30 days (responses ranged from "strongly agree" to "strongly disagree"); patients were classified as nonadherent if they responded "strongly agree," "agree," "don't know," or "refused" for any of the four statements.

ACDC, Advanced Comprehensive Diabetes Care; DBP, diastolic blood pressure; HbA1c, hemoglobin A1c; PHQ-9, Patient Health Questionnaire-9; REALM, Rapid Estimate of Adult Literacy in Medicine; SBP, systolic blood pressure; SCI-R, Self-Care Inventory–Revised; SD, standard deviation.

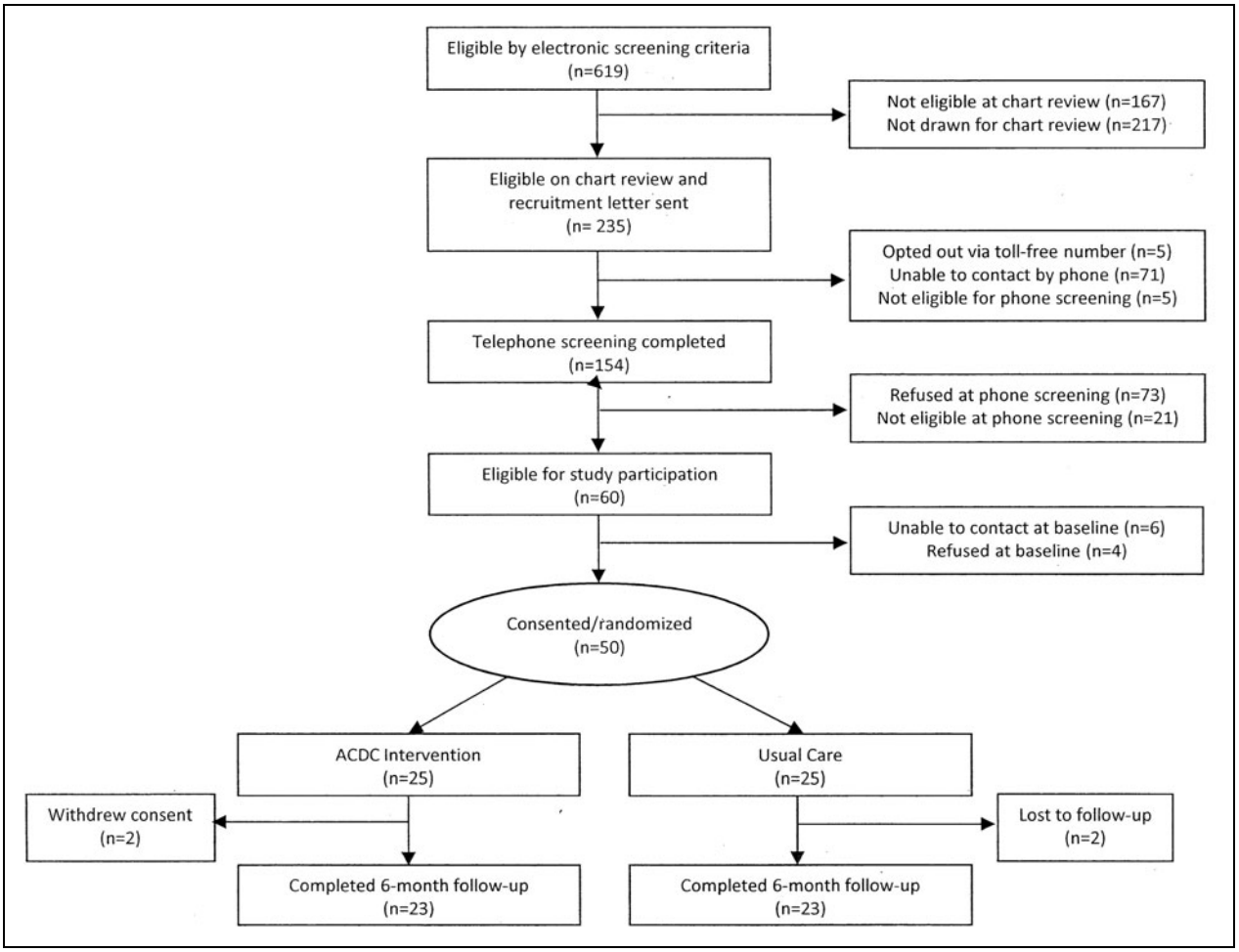

Fig. 1. Study Consolidated Standards of Reporting Trials diagram. ACDC, Advanced Comprehensive Diabetes Care.

<5 min per intervention encounter. Based on the four TAP measure components, participants found ACDC acceptable (*Table 4*). In particular, participants felt that ACDC was an effective way to address their diabetes and expressed willingness to continue applying the skills they had learned.

ADVERSE EVENTS

Adverse events were uncommon and similar between groups. No participants reported hypoglycemia requiring assistance. Among the 20 ACDC participants transmitting SMBG data during the study, hypoglycemia was most common during the period after enrollment but prior to the first intervention telephone encounter. Hypoglycemia was rare thereafter, with participants reporting a median of zero SMBG values <70 mg/dL (interquartile range, 0–4) and zero <60 mg/dL (interquartile range, 0–1). Three veterans had more than five blood glucose values <70 mg/dL after intervention initiation, and only one had more than five such values <60 mg/dL; this veteran was identified as having HbA1c–blood glucose discordance, and his hypoglycemia resolved by study end.

Discussion

In order to reduce the burden of PPDM, there is a need for comprehensive telemedicine interventions that can be practically and effectively implemented in real-world settings. This pilot study demonstrated that existing VHA clinical staff can deliver comprehensive telemedicine-based care for PPDM using standard VHA equipment. ACDC appeared to reduce HbA1c versus usual care among veterans whose diabetes had remained persistently poorly controlled despite clinic-based care. Given ACDC's potential for scalability and effectiveness, this approach may be a vital step toward reducing complications and costs in the clinic-refractory PPDM population, where such improvements are critically needed.

 © MARY ANN LIEBERT, INC.

Table 2. Results of Longitudinal Data Models Comparing Study Outcomes in Intervention Versus Usual Care Patients

OUTCOME	PREDICTED MEAN (SE)		PREDICTED MEAN DIFFERENCE ACDC VERSUS UC (95% CI)	*P* VALUE
	ACDC INTERVENTION	UC		
HbA1c (%)[a]				
Baseline	10.5 (0.2)	10.5 (0.2)		
3 months	9.2 (0.3)	10.2 (0.3)	−1.0 (−1.7, −0.2)	0.012
6 months	9.2 (0.4)	10.2 (0.4)	−1.0 (−2.0, −0.0)	0.050
SBP (mm Hg)[b]				
Baseline	126.7 (2.3)	126.7 (2.3)		
3 months	126.4 (3.1)	128.7 (3.1)	−2.3 (−10.4, 5.9)	0.575
6 months	122.1 (3.0)	129.8 (2.7)	−7.7 (−14.8, −0.6)	0.035
DBP (mm Hg)[b]				
Baseline	72.3 (1.5)	72.3 (1.5)		
3 months	70.7 (1.9)	72.4 (1.9)	−1.7 (−6.6, 3.2)	0.498
6 months	67.8 (1.9)	73.4 (1.8)	−5.6 (−9.9, −1.2)	0.013
SCI-R[c]				
Baseline	49.3 (1.7)	49.3 (1.7)		
3 months	54.7 (2.4)	52.4 (2.3)	2.3 (−3.4, 8.0)	0.421
6 months	58.0 (2.7)	50.9 (2.6)	7.0 (0.1, 14.0)	0.047
PHQ-9[c]				
Baseline	7.1 (0.8)	7.1 (0.8)		
3 months	5.4 (1.0)	6.4 (1.0)	−0.9 (−3.3, 1.4)	0.428
6 months	6.6 (1.0)	5.5 (1.0)	1.1 (−1.4, 3.6)	0.397
	PREDICTED PROBABILITY (95% CI)		ODDS RATIO ACDC INTERVENTION VERSUS UC (95% CI)	
	ACDC INTERVENTION	UC		
Self-reported medication nonadherence[c]				
Baseline	0.64 (0.50, 0.76)	0.64 (0.50, 0.76)		
3 months	0.41 (0.25, 0.59)	0.44 (0.26, 0.62)	0.90 (0.35, 2.34)	0.830
6 months	0.36 (0.20, 0.56)	0.35 (0.19, 0.56)	1.02 (0.33, 3.19)	0.970

[a]Primary analysis. Models include data from 50 patients ($n=25$ in the intervention [Advanced Comprehensive Diabetes Care (ACDC)] group and $n=25$ in the usual care [UC] group).

[b]Values represent means of three measurements performed by research staff at study visit. Models include data from 49 patients ($n=25$ in the intervention group and $n=24$ in the UC group).

[c]Models include data from 50 patients ($n=25$ in the intervention group and $n=25$ in the UC group).

CI, confidence interval; DBP, diastolic blood pressure; HbA1c, hemoglobin A1c; PHQ-9, Patient Health Questionnaire-9; SBP, systolic blood pressure; SCI-R, Self-Care Inventory-Revised; SE, standard error.

IMPLICATIONS OF FINDINGS

We designed ACDC to be effective for PPDM by combining four evidence-based strategies that target factors known to underlie poor diabetes control. We found that ACDC improved HbA1c by 1.0%; if sustained, this change could reduce microvascular complications by 37%, myocardial infarction by 14%, and diabetes-related death by 21% for participants.[1] It is interesting that HbA1c improved by 1.0% after 3 months but then appeared to plateau from 3 to 6 months. Although this may seem to suggest that ACDC had less impact after 3 months, we found that engaged participants (six or more encounters) did continue to improve between 3 and 6 months. Overall, participants with high intervention engagement responded especially well to ACDC, with observed HbA1c improving by 1.9%.

By contrast, veterans with lower engagement (three or fewer encounters) fared less well. Because factors that underlie PPDM, such as competing demands from employment and suboptimal self-management,[7] may likewise affect general participation with medical care, we anticipated challenges engaging participants. Our Consolidated Standards of Reporting Trials diagram (*Fig. 1*) indicated that 73 of 154 patients (47%) approached for telephone screening declined participation. This number may have been lower had we compensated participants and may decrease when ACDC is presented as clinical care rather than research. It is encouraging that, once enrolled, most intervention participants successfully engaged ACDC, found the intervention acceptable based on the TAP measure, and exhibited clinical improvement.

ACDC is an intensive approach, and refinement for further study and eventual practical use will be crucial. As part of our ongoing qualitative work, we will explore if and how each intervention component benefitted participants. Enhanced medication management was likely important for many veterans, and their improvements in diabetes self-care suggest that self-management support was likewise helpful. By contrast, ACDC did not impact depression, which may relate

Table 3. Change in Observed Hemoglobin A1c During the Study Period for Engaged (Six or More Study Encounters) Versus Nonengaged Intervention Patients (Three or Fewer Study Encounters)

	ENGAGED PATIENTS (≥6 STUDY ENCOUNTERS)			NONENGAGED PATIENTS (≤3 STUDY ENCOUNTERS)			USUAL CARE		
HBA1C (%)	*N*	MEAN	SD	*N*	MEAN	SD	*N*	MEAN	SD
Baseline	16	10.2	(1.2)	9	10.3	(1.7)	24	10.7	(1.6)
3 months	16	8.7	(0.9)	7	10.2	(1.6)	24	10.3	(1.8)
6 months	16	8.3	(1.0)	7	10.8	(2.8)	23	10.4	(1.6)

HbA1c, hemoglobin A1c; SD, standard deviation.

to high baseline utilization of psychiatry services within VHA. Our depression component thus requires re-evaluation. As we continue to refine ACDC, we will incorporate a maintenance strategy to help patients retain improvements in glycemic control.

Although ACDC did not directly address hypertension, BP improved in the intervention group relative to usual care. Our future work will further explore this finding.

FINDINGS IN THE CONTEXT OF CURRENT LITERATURE

This work is innovative in two ways. First, we focused exclusively on veterans with clinic-refractory PPDM. Because complications and costs rise as HbA1c increases,[1–3] patients with PPDM contribute disproportionately to the nationwide diabetes burden,[35] and so are a priority target for intervention. Even in this refractory population, ACDC had an estimated effect of 1.0%, exceeding that of many published telemedicine interventions for diabetes.[11–13] ACDC's apparent potency may relate to its bundling of multiple evidence-based approaches. Furthermore, our cohort's markedly elevated baseline HbA1c provided ample opportunity for improvement as compared with better-controlled populations.

Second, in order to foster comprehensive telemedicine-based management as a real-world option for PPDM, we explicitly designed ACDC for practical delivery using existing clinical VHA staffing and equipment. ACDC repurposes telemedicine infrastructure that is ubiquitous at VHA centers nationwide, which gives this approach unique potential scalability. By leveraging VHA's investment in telemedicine infrastructure, ACDC has a natural avenue for dissemination throughout VHA and therefore substantial promise to widely impact PPDM. Our ongoing work will hone ACDC for broader study and clinical use.

Care delivery technology evolves continually, and for some patients intervention delivery using Web or smartphone platforms may be preferable to telephone. However, the potential advantages of leading-edge technology must be weighed against the potential for implementation. Given current VHA infrastructure, telephone-based care strikes a reasonable balance among effectiveness, reach, and potential scalability. As VHA increasingly implements alternatives to telephone-based care delivery, ACDC will evolve accordingly.

Table 4. Treatment Acceptability and Preferences Measure for Intervention Patients

TAP MEASURE ITEM[a]	*N*[b]	MEAN (SD)
ACDC seemed like an appropriate way to address my diabetes.	22	3.27 (1.35)
ACDC fit my individual lifestyle.	21	3.05 (1.36)
ACDC was an effective way to address my diabetes.	22	3.36 (1.40)
I am willing to apply the skills I learned in ACDC and adhere to the program.	22	3.50 (1.22)

[a]Using a 5-point Likert scale, patients addressed the extent to which four statements described their perceptions of the ACDC intervention; responses ranged from 0 ("not at all") to 4 ("very much").

[b]One patient responded "don't know" to all items, and an additional patient responded "don't know" to Item 2.

ACDC, Advanced Comprehensive Diabetes Care; SD, standard deviation; TAP, Treatment Acceptability and Preferences.

LIMITATIONS

Along with limitations discussed above, this was a single-center pilot study focusing on a small, primarily male, veteran population with insulin-requiring type 2 diabetes. Our findings should be interpreted accordingly. We used two HT nurses and a single physician for intervention implementation, all of whom were part of the study team; future work will test ACDC at different sites with local intervention teams to assure generalizability. Because ACDC is designed for implementation using VHA-specific infrastructure, our findings do not immediately translate to other settings.

This study was not specifically designed to examine effectiveness; although our results suggest that ACDC may be effective, future work is needed to confirm these findings. A formal cost-effectiveness assessment will also be needed to inform wider implementation.

© MARY ANN LIEBERT, INC.

Conclusions

We have demonstrated that existing VHA clinical staffing can deliver comprehensive, telemedicine-based care for PPDM using standard VHA equipment. ACDC's practicality and effectiveness may position it to meaningfully reduce the burden of diabetes within VHA, the largest integrated healthcare system in the United States. Extending our evaluation of the ACDC approach beyond VHA is critical. Given recent increases in reimbursement for chronic disease management,[36] the time is right to promote broad scaling of practical telemedicine-based disease management approaches for PPDM and other chronic illnesses that do not respond to clinic-based care.

Acknowledgments

We gratefully acknowledge contributions made by Stephanie Macy, research assistant at the Durham VAMC Center for Health Services Research in Primary Care. ACDC was supported by grant RRP 12-458 from the Veterans Affairs Diabetes QUERI. M.J.C. is supported by Career Development Award 13-261 from the Veterans Affairs Health Services Research and Development. H.B.B. is supported by Research Career Scientist award RCS 08-027 from the Veterans Affairs Health Service Research and Development.

Disclosure Statement

No competing financial interests exist.

REFERENCES

1. Stratton IM, Adler AI, Neil HA, et al. Association of glycaemia with macrovascular and microvascular complications of type 2 diabetes (UKPDS 35): Prospective observational study. *BMJ* **2000;**321:405–412.
2. Gilmer TP, O'Connor PJ, Manning WG, Rush WA. The cost to health plans of poor glycemic control. *Diabetes Care* **1997;**20:1847–1853.
3. McBrien KA, Manns BJ, Chui B, et al. Health care costs in people with diabetes and their association with glycemic control and kidney function. *Diabetes Care* **2013;**36:1172–1180.
4. Ali MK, Bullard KM, Saaddine JB, Cowie CC, Imperatore G, Gregg EW. Achievement of goals in U.S. diabetes care, 1999–2010. *N Engl J Med* **2013;**368:1613–1624.
5. Ali MK, Bullard KM, Imperatore G, Barker L, Gregg EW. Characteristics associated with poor glycemic control among adults with self-reported diagnosed diabetes—National Health and Nutrition Examination Survey, 2007–2010. *MMWR Surveill Summ* **2012;**61:32–37.
6. Davidson J. Strategies for improving glycemic control: Effective use of glucose monitoring. *Am J Med* **2005;**118(Suppl 9A):27S–32S.
7. Crowley MJ, Holleman R, Klamerus ML, Bosworth HB, Edelman D, Heisler M. Factors associated with persistent poorly-controlled diabetes mellitus (PPDM): Clues to improving management in patients with resistant poor control. *Chronic Illn* **2014;**10:291.
8. Richardson LK, Egede LE, Mueller M, Echols CL, Gebregziabher M. Longitudinal effects of depression on glycemic control in veterans with Type 2 diabetes. *Gen Hosp Psychiatry* **2008;**30:509–514.
9. Morrison F, Shubina M, Turchin A. Encounter frequency and serum glucose level, blood pressure, and cholesterol level control in patients with diabetes mellitus. *Arch Intern Med* **2011;**171:1542–1550.
10. American Telemedicine Association. What is telemedicine? Available at www.americantelemed.org/about-telemedicine/what-is-telemedicine (last accessed July 26, 2015).
11. Health Quality Ontario. Home telemonitoring for type 2 diabetes: An evidence-based analysis. *Ont Health Technol Assess Ser* **2009;**9(24):1–38.
12. Health Quality Ontario. Behavioural interventions for type 2 diabetes: An evidence-based analysis. *Ont Health Technol Assess Ser* **2009;**9(21):1–45.
13. Pimouguet C, Le Goff M, Thiébaut R, Dartigues JF, Helmer C. Effectiveness of disease-management programs for improving diabetes care: A meta-analysis. *CMAJ* **2011;**183:E115–E127.
14. Aburizik A, Dindo L, Kaboli P, Charlton M, Dawn K, Turvey C. A pilot randomized controlled trial of a depression and disease management program delivered by phone. *J Affect Disord* **2013;**151:769–774.
15. U.S. Department of Veterans Affairs. VA telehealth services. Available at www.telehealth.va.gov/index.asp (last accessed July 26, 2015).
16. Glasgow RE, Lichtenstein E, Marcus AC. Why don't we see more translation of health promotion research to practice? Rethinking the efficacy-to-effectiveness transition. *Am J Public Health* **2003;**93:1261–1267.
17. Milat AJ, King L, Bauman AE, Redman S. The concept of scalability: Increasing the scale and potential adoption of health promotion interventions into policy and practice. *Health Promot Int* **2013;**28:285–298.
18. Zullig LL, Gellad WF, Moaddeb J, et al. Improving diabetes medication adherence: Successful, scalable interventions. *Patient Prefer Adherence* **2015;**9:139–149.
19. U.S. Department of Veterans Affairs. Veterans Health Administration. Available at www.va.gov/health (last accessed August 20, 2015).
20. Crowley MJ, Bosworth HB, Coffman CJ, et al. Tailored Case Management for Diabetes and Hypertension (TEACH-DM) in a community population: Study design and baseline sample characteristics. *Contemp Clin Trials* **2013;**36:298–306.
21. American Diabetes Association. Standards of medical care in diabetes—2015: Glycemic targets. *Diabetes Care* **2015;**38(Suppl 1):S33–S40.
22. Lustman PJ, Williams MM, Sayuk GS, Nix BD, Clouse RE. Factors influencing glycemic control in type 2 diabetes during acute- and maintenance-phase treatment of major depressive disorder with bupropion. *Diabetes Care* **2007;**30:459–466.
23. Weinger K, Butler HA, Welch GW, La Greca AM. Measuring diabetes self-care: A psychometric analysis of the Self-Care Inventory-Revised with adults. *Diabetes Care* **2005;**28:1346–1352.
24. Arroll B, Goodyear-Smith F, Crengle S, et al. Validation of PHQ-2 and PHQ-9 to screen for major depression in the primary care population. *Ann Fam Med* **2010;**8:348–353.
25. Morisky DE, Green LW, Levine DM. Concurrent and predictive validity of a self-reported measure of medication adherence. *Med Care* **1986;**24:67–74.
26. Bosworth HB, Olsen MK, Neary A, et al. Take Control of Your Blood pressure (TCYB) study: A multifactorial tailored behavioral and educational intervention for achieving blood pressure control. *Patient Educ Couns* **2008;**70:338–347.
27. Bent S, Padula A, Avins AL. Brief communication: Better ways to question patients about adverse medical events: A randomized, controlled trial. *Ann Intern Med* **2006;**144:257–261.
28. Sidani S, Epstein DR, Bootzin RR, Moritz P, Miranda J. Assessment of preferences for treatment: Validation of a measure. *Res Nurs Health* **2009;**32:419–431.
29. Edelman D, Fredrickson SK, Melnyk SD, et al. Medical clinics versus usual care for patients with both diabetes and hypertension: A randomized trial. *Ann Intern Med* **2010;**152:689–696.

30. Hulley SB, Cummings SR, Browner WS, Grady DG, Newman TB. *Designing clinical research*, 3rd ed. Philadelphia: Lippincott, Williams & Wilkins, **2007.**

31. Verbeke G, Molenberghs G. *Linear mixed models for longitudinal data*. New York: Springer-Verlag, **2000.**

32. Fitzmaurice G, Laird N, Ware J. *Applied longitudinal analysis*. Hoboken, NJ: Wiley, **2004.**

33. Little RJ, Rubin DB. *Statistical analysis with missing data*. Hoboken, NJ: Wiley, **2002.**

34. Diggle PJ, Heagerty P, Liang KY, Zeger SL. *Analysis of longitudinal data*. Oxford, United Kingdom: Oxford University Press, **2002.**

35. American Diabetes Association. Economic costs of diabetes in the U.S. in 2012. *Diabetes Care* **2013;**36:1033–1046.

36. Edwards ST, Landon BE. Medicare's chronic care management payment—Payment reform for primary care. *N Engl J Med* **2014;**371: 2049–2051.

Address correspondence to:
Matthew J. Crowley, MD
HSR&D (152)
Durham Veterans Affairs Medical Center
508 Fulton Street
Durham, NC 27705

E-mail: matthew.crowley@dm.duke.edu

Received: July 26, 2015
Revised: August 24, 2015
Accepted: August 26, 2015

© MARY ANN LIEBERT, INC.